

Unit 1 Physics on the go	65
Unit 2 Physics at Work	73
Unit 3 Exploring Physics	83
Unit 4 Physics on the Move	91
Unit 5 Physics from Creation to Collapse	101
Unit 6 Experimental Physics	109

Course structure

- Edexcel's GCE in Physics comprises six units and contains an Advanced Subsidiary subset of three AS units.
- The Advanced Subsidiary GCE is the first half of the GCE course and consists of Units 1, 2 and 3. It may be awarded as a discrete qualification or can contribute 50 per cent of the total Advanced GCE marks.
- The full Advanced GCE award consists of the three AS units (Units 1, 2 and 3), plus three A2 units (Units 4, 5 and 6) which make up the other 50 per cent of the Advanced GCE. Students wishing to take the full Advanced GCE must, therefore, complete all six units.
- The structure of this qualification allows teachers to construct a course of study which can be taught and assessed either as:
 - ◆ distinct modules of teaching and learning with related units of assessment taken at appropriate stages during the course; or
 - ◆ a linear course which is assessed in its entirety at the end.

Administration of internal assessment

Internal standardisation

Teachers choosing the option of marking the internal assessment must show clearly how the marks have been awarded in relation to the assessment criteria. If more than one teacher in a centre is marking students' work, there must be a process of internal standardisation to ensure that there is consistent application of the assessment criteria.

Authentication

All candidates must sign an authentication statement. Statements relating to work not sampled should be held securely in your centre. Those which relate to sampled candidates must be attached to the work and sent to the moderator. In accordance with a revision to the current Code of Practice, any candidate unable to provide an authentication statement will receive zero credit for the component. Where credit has been awarded by a centre-assessor to sampled work without an accompanying authentication statement, the moderator will inform Edexcel and the mark will be adjusted to zero.

**Further
information**

For more information on annotation, authentication, mark submission and moderation procedures, please refer to the *Edexcel AS and GCE in Physics: Instructions and administrative documentation for internally assessed units* document, which is available on the Edexcel website.

For up-to-date advice on teacher involvement, please refer to the Joint Council for Qualifications (JCQ) — Instructions for conducting coursework/portfolio document on the JCQ website: www.jcq.org.uk. For up-to-date advice on malpractice and plagiarism, please refer to the Joint Council for Qualifications — Suspected Malpractice in Examinations: Policies and Procedures and the Joint Council for Qualifications (JCQ) — Instructions for conducting coursework/portfolio documents on the JCQ website: www.jcq.org.uk.

CONCEPT-LED APPROACH

The following section shows how the specification may be taught using the concept-led approach

1.1 Introduction

Concept approach

This unit covers mechanics and materials. The unit may be taught using either a concept approach or a context approach. The concept approach begins with a study of the laws, theories and models of physics and then explores their practical applications. This section of the specification is presented in a format for teachers who wish to use the concept approach.

Context approach

This unit is presented in a different format on page 65 for teachers who wish to use a context approach. The context approach begins with the consideration of an application that draws on many different areas of physics, and then the laws, theories and models of physics that apply to this application are studied. The context approach for this unit uses three contexts for teaching this unit: sports, the production of sweets and biscuits and spare part surgery.

How Science Works

The QCA's GCE Science Criteria includes *How Science Works* (see *Appendix 6*). This should be integrated with the teaching and learning of this unit.

It is expected that students will be given opportunities to use spreadsheets and computer models to analyse and present data, and make predictions.

The word 'investigate' indicates where students should develop their practical skills for *How Science Works*, numbers 1–6 as detailed in *Appendix 6* (internal assessment may require these skills). Students should communicate the outcomes of their investigations using appropriate scientific, technical and mathematical language, conventions and symbols.

Applications of physics should be studied using a range of contemporary contexts that relate to this unit.